

Predicting High-Income Individuals for phiAI Donor Outreach: A Logistic Regression

Approach

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Overview

This report documents a supervised binary classification model built to predict whether an individual's annual income exceeds \$50,000, using demographic and employment data from the 1994 U.S. Census. The model was developed for phiAI, an EdTech-focused nonprofit, to help prioritize donor outreach for its STEM scholarship programs by identifying individuals whose profile suggests high earning potential. The dataset used is the UCI Adult / Census Income dataset (Kohavi, 1996), sourced from the UCI Machine Learning Repository (Kelly et al., n.d.) at <https://archive.ics.uci.edu/dataset/2/adult>. A Logistic Regression classifier, tuned via grid search, was selected as the final model after comparison against a Decision Tree and a Support Vector Classifier.

Dataset Description

The dataset represents individual-level demographic and employment records drawn from the 1994 U.S. Census Bureau's Current Population Survey (Kohavi, 1996). After the data-quality review described in the accompanying notebook, the working dataset contains 45,222 rows and 14 columns: 13 candidate features plus the binary target variable, income (whether an individual earns more than \$50,000 per year). Features span both continuous variables (age, education-num, capital-gain, capital-loss, hours-per-week) and categorical variables (workclass, education_level, marital-status, occupation, relationship, race, sex, native-country). After one-hot encoding of the categorical variables, the modeling feature space expands to 103 columns. The

target class is imbalanced: only 24.78% of individuals in the dataset earn more than \$50,000 annually.

Modeling Approach

Data preparation followed standard practice for mixed categorical/continuous tabular data. The two most heavily skewed continuous features, capital-gain and capital-loss, were log-transformed to reduce the influence of extreme outliers, a common remedy for right-skewed financial variables (Hastie et al., 2009). All continuous features were then scaled to the $[0, 1]$ range using min-max normalization, and categorical features were one-hot encoded into binary indicator columns, both standard preprocessing steps for models that are sensitive to feature scale and require numeric input (Hastie et al., 2009; Pedregosa et al., 2011).

Three candidate models were trained and compared: a Decision Tree, a Support Vector Classifier (SVC) with an RBF kernel (Cortes & Vapnik, 1995), and a Logistic Regression classifier (Hosmer et al., 2013). Logistic Regression was ultimately selected because it tied the SVC on test accuracy and F-beta score while training roughly 274x faster (142.28 seconds versus 0.52 seconds on the full training set), and because the feature space, 103 columns of which roughly 98 are sparse one-hot indicators, is exactly the kind of high-dimensional, largely linear setting where kernel methods such as SVC's RBF kernel offer little advantage over a linear model (Cortes & Vapnik, 1995). Hyperparameters (regularization strength C and the $L1/L2$ balance) were tuned using grid search cross-validation (Bergstra & Bengio, 2012).

The evaluation metric was F-beta score with $\beta = 0.5$, which weights precision above recall (van Rijsbergen, 1979). This choice reflects phiAI's business priority: incorrectly flagging a low-income individual as a likely donor wastes outreach effort, so false positives are more costly than false negatives, making a precision-weighted metric more appropriate than plain

accuracy or an equally weighted F1 score, particularly given the dataset's class imbalance (Hastie et al., 2009). Logistic Regression's core assumption is that the log-odds of the outcome are a linear combination of the input features (Hosmer et al., 2013); this is a simplifying assumption that will not perfectly capture any true non-linear interactions between features such as education and occupation.

Results

On the full test set, the naive baseline (always predicting income > \$50,000) achieved 0.2478 accuracy and 0.2917 F-beta score, as shown in the Naive Predictor Benchmark section of the notebook. The untuned Logistic Regression model reached 0.8304 accuracy and 0.6577 F-beta score in the Initial Model Comparison table, and grid search tuning (selecting C=1 with an L1 penalty) improved this marginally to 0.8306 accuracy and 0.6581 F-beta score, per the Tuned Model Results table. This represents a substantial improvement over the naive baseline and is statistically indistinguishable from the SVC model's 0.8303 accuracy / 0.6573 F-beta score, while training in a fraction of the time. Feature importance analysis (Extracting Feature Importance section, using a separately trained Decision Tree's `feature_importances_` attribute) identified marital status (being married), Bachelor's-level education, and executive/managerial or professional-specialty occupation as the five most predictive features. As shown in the Effect of Feature Selection section, a reduced model using only these five features retained most of the predictive power (0.8192 accuracy, 0.6302 F-beta score) but showed a meaningful F-beta drop of about 4.2% relative to the full model, indicating the remaining ~98 features still contribute meaningfully to identifying likely high earners.

Interpretation for a Non-Technical Audience

In plain terms, this model looks at a person's background, things like their job type, education level, marital status, and how many hours they work, and estimates how likely they are to earn more than \$50,000 a year. Out of every 100 people the model reviews, roughly 83 are correctly categorized as above or below that income threshold. The model is not perfect: it will occasionally flag someone as a likely high earner when they are not, and occasionally miss someone who is. It was specifically tuned to make the first kind of mistake less often than the second, because for phiAI, wasting outreach effort on someone who is not a strong donor prospect is more costly than not reaching out to every single high earner. The single strongest signal the model found was marital status, specifically, being married, followed by having a Bachelor's degree and working in a managerial or specialized professional role.

Limitations and Potential Bias

Limitation. The most significant limitation of this model is its age: the underlying data was collected in 1994, and the economic relationships it captures, income by occupation, education, and demographic group, are three decades out of date. A model trained on this data would not reliably predict income for people today without retraining on more recent data.

Potential bias. The most significant source of potential bias is the inclusion of sex, race, and marital-status as raw predictive features. Historical income data reflects real, documented inequities in pay and occupational access along these lines (Ding et al., 2021), and models trained directly on such features risk learning and reproducing those disparities rather than identifying donor potential fairly, for example, systematically under-flagging qualified individuals from groups that were historically paid less for comparable work. As a mitigation step, feature-importance analysis confirmed that sex and race were not among the model's five most predictive features, and a further mitigation, excluding those two features entirely and re-

evaluating performance, is proposed as a direct next step before this model is used in any real outreach decision (Ding et al., 2021).

References

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